

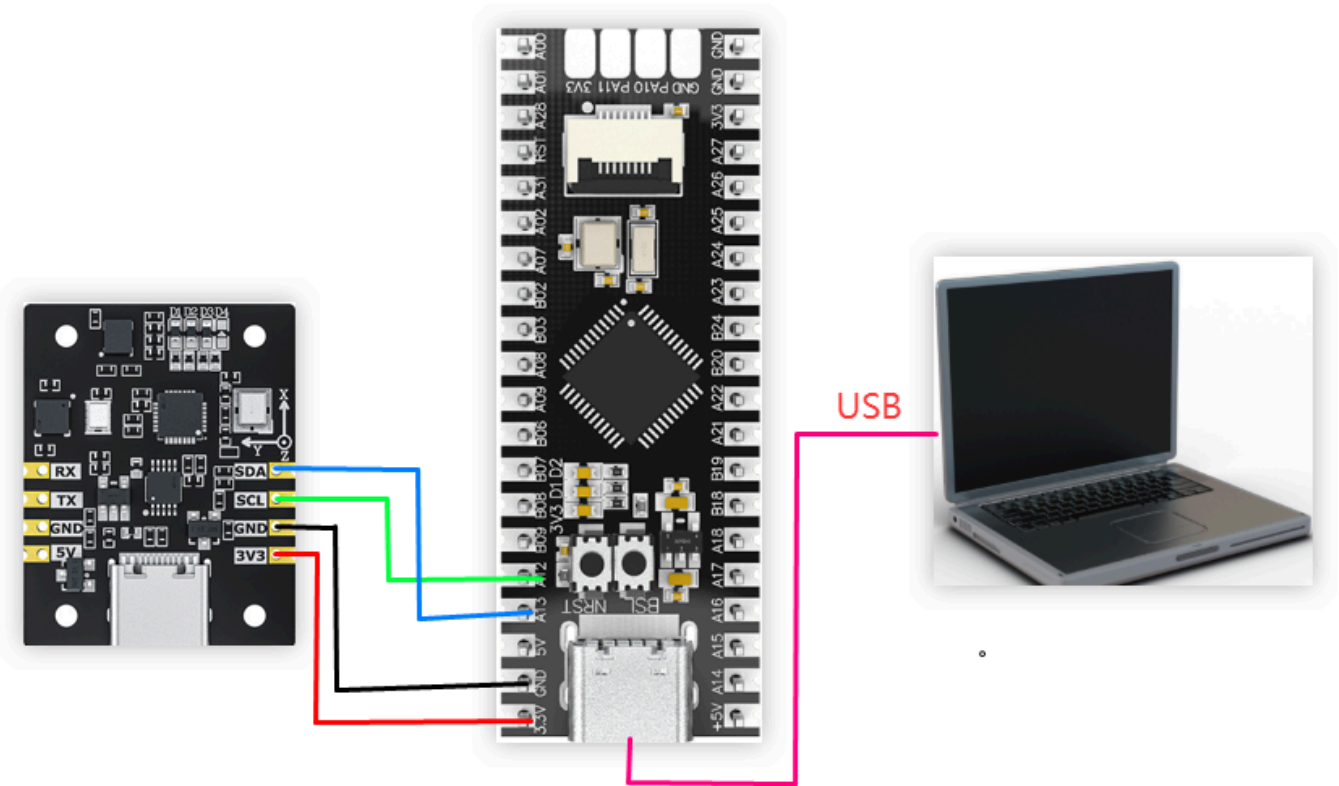
MSPM0-IIC Data Reading

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- 1. Connecting the Device
- 2. Key Code Analysis
- 3. Reading IMU Data

This example uses the MSPM0G3519 core board (Yabo), a Windows computer, several DuPont wires, and an IMU attitude sensor.

1. Connecting the Device



IMU Attitude Sensor	MSPM0G3519 Core Board (Yahboom)
SDA	A13
SCL	A12
GND	GND
3V3	3V3

2. Key Code Analysis

Please refer to the source code in the documentation for specific code.

```
/**
 * @brief Read acceleration in g.
 *
 */
int IMU_I2C_ReadAccelerometer(float out[3])
{
    uint8_t register_data[6];
    if (read_register(IMU_FUNC_RAW_ACCEL, register_data, 6) != 0) {
        return -1;
    }
    if (out != NULL) {
        float ratio = 16.0f / 32767.0f;
        out[0] = to_int16(&register_data[0]) * ratio;
        out[1] = to_int16(&register_data[2]) * ratio;
        out[2] = to_int16(&register_data[4]) * ratio;
    }
    return 0;
}

/**
 * @brief Read angular velocity in rad/s.
 *
 */
int IMU_I2C_ReadGyroscope(float out[3])
{
    uint8_t register_data[6];
    if (read_register(IMU_FUNC_RAW_GYRO, register_data, 6) != 0) {
        return -1;
    }
    if (out != NULL) {
        float ratio = (2000.0f / 32767.0f) * (3.1415926f / 180.0f);
        out[0] = to_int16(&register_data[0]) * ratio;
        out[1] = to_int16(&register_data[2]) * ratio;
        out[2] = to_int16(&register_data[4]) * ratio;
    }
    return 0;
}

/**
 * @brief Read magnetic field strength in micro tesla.
 *
 */
int IMU_I2C_ReadMagnetometer(float out[3])
{
    uint8_t register_data[6];
    if (read_register(IMU_FUNC_RAW_MAG, register_data, 6) != 0) {
        return -1;
    }
    if (out != NULL) {
```

```

        float ratio = 800.0f / 32767.0f;
        out[0] = to_int16(&register_data[0]) * ratio;
        out[1] = to_int16(&register_data[2]) * ratio;
        out[2] = to_int16(&register_data[4]) * ratio;
    }
    return 0;
}

/**
 * @brief Read quaternion (w, x, y, z).
 *
 */
int IMU_I2C_ReadQuaternion(float out[4])
{
    uint8_t register_data[16];
    if (read_register(IMU_FUNC_QUAT, register_data, 16) != 0) {
        return -1;
    }
    if (out != NULL) {
        out[0] = to_float(&register_data[0]);
        out[1] = to_float(&register_data[4]);
        out[2] = to_float(&register_data[8]);
        out[3] = to_float(&register_data[12]);
    }
    return 0;
}

/**
 * @brief Read Euler angles (rad).
 *
 */
int IMU_I2C_ReadEuler(float out[3])
{
    uint8_t register_data[12];
    if (read_register(IMU_FUNC_EULER, register_data, 12) != 0) {
        return -1;
    }
    if (out != NULL) {
        const float RAD2DEG = 57.2957795f;
        out[0] = to_float(&register_data[0]) * RAD2DEG;
        out[1] = to_float(&register_data[4]) * RAD2DEG;
        out[2] = to_float(&register_data[8]) * RAD2DEG;
    }
    return 0;
}

```

IMU_I2C_ReadAccelerometer(): Reads acceleration data (in g)

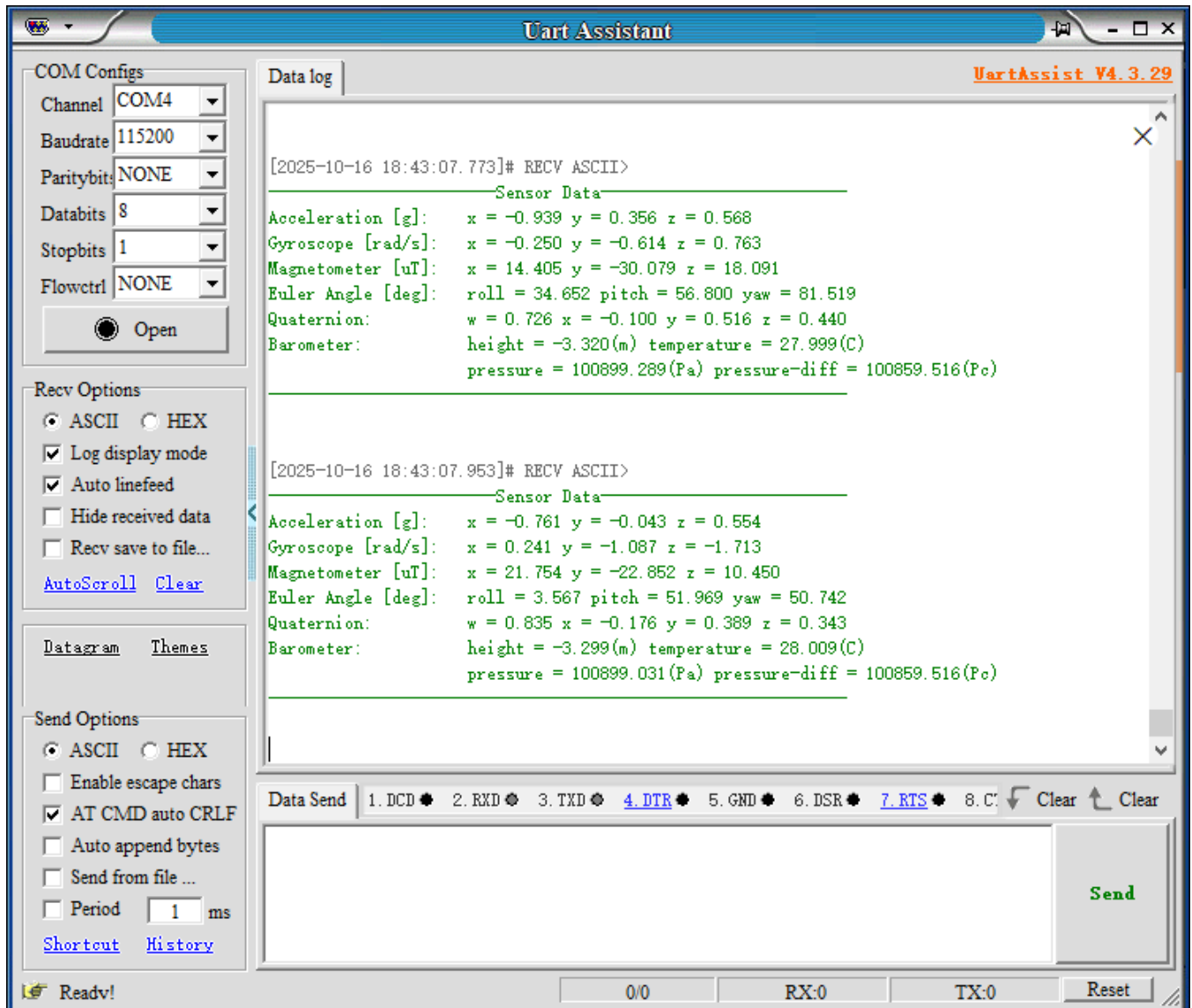
IMU_I2C_ReadGyroscope(): Reads angular velocity (in g) (rad/s)

IMU_I2C_ReadQuaternion(): Reads quaternions

IMU_I2C_ReadEuler(): Reads Euler angles (radians)

3. Reading IMU Data

After the program is downloaded and enters MSPM0, open the serial port assistant (configuration parameters are shown in the figure below). You can see the IMU module's data being continuously printed. When we change the IMU module's attitude, the data will change.



Note: The above is the data reading for a 10-axis IMU, 6-axis without magnetometer and barometer data, and 9-axis without barometer data.